Chapter test

23 Personal health and infectious diseases

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** an infectious disease?

   A. hepatitis B
   B. measles
   C. cholera
   D. lactose intolerance ☐

★ 2. Which of the following combinations is correct?

	Disease	*Causative agent*	*Preventive measure*
A.	AIDS	virus	take contraceptive pill before sexual intercourse
B.	cholera	bacterium	avoid direct contact with infected people
C.	malaria	protist	apply mosquito repellent
D.	influenza	bacterium	wear a surgical face mask if infected ☐

3. What is the role of mosquitoes in the transmission of dengue fever?

   A. pathogen
   B. host
   C. parasite
   D. vector ☐

4. Which of the following diseases is/are caused by viruses?

   (1) Zika fever
   (2) tuberculosis
   (3) chickenpox

   A. (1) only
   B. (2) and (3) only
   C. (1) and (3) only
   D. (1), (2) and (3) ☐

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★ 5. Which of the following contraception method is the most effective in preventing the spread of syphilis?

A. diaphragms
B. male condoms
C. spermicide
D. vasectomy ☐

6. Which of the following are appropriate preventive measures against hepatitis B?

(1) Do not share needles and syringes for any injections.
(2) Use condom during sexual intercourse.
(3) Avoid direct physical contact with infected people.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

✂ 7. Which of the following statements about antibiotics is/are correct?

(1) Antibiotics can treat all infectious diseases.
(2) Some antibiotics inhibit the synthesis of bacterial cell walls.
(3) One type of antibiotics can only kill one type of microorganisms.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only ☐

✂★ 8. After starting a course of antibiotics, the patient may still feel ill for two or three more days as

A. pathogens are still multiplying in the body.
B. antibiotics cause damage to both pathogens and body tissues.
C. antibiotics can only relieve the symptoms.
D. antibiotics start to kill pathogens after the whole course is completed. ☐

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✂★ 9. Why do scientists continue to search for new antibiotics even many antibiotics are already available?

A. Antibiotics lost their functions over time.
B. Antibiotic-resistant bacteria have evolved.
C. The treatment of viral infections requires a broad range of antibiotics.
D. Broad-spectrum antibiotics are more effective in killing pathogens. ☐

✂ 10. Which of the following about sulpha drugs is/are ***incorrect***?

(1) Sulpha drugs are synthetic chemical compounds.
(2) Sulpha drugs are effective against viral diseases.
(3) Sulpha drugs are enzyme inhibitors.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	23.2	23.4	23.4	23.3	23.4	23.4	23.5	23.5	23.5	23.5

B. Problem solving (19 marks)

★ 11. A person was contracted with typhoid (傷寒). Typhoid is caused by the bacterium *Salmonella typhi*. It is transmitted by contaminated water or food. The symptoms include a severe fever and diarrhoea.

(a) Explain why the following symptoms of typhoid may lead to death if the person did not receive proper treatment.

(i) high fever (3 marks)

(ii) diarrhoea (2 marks)

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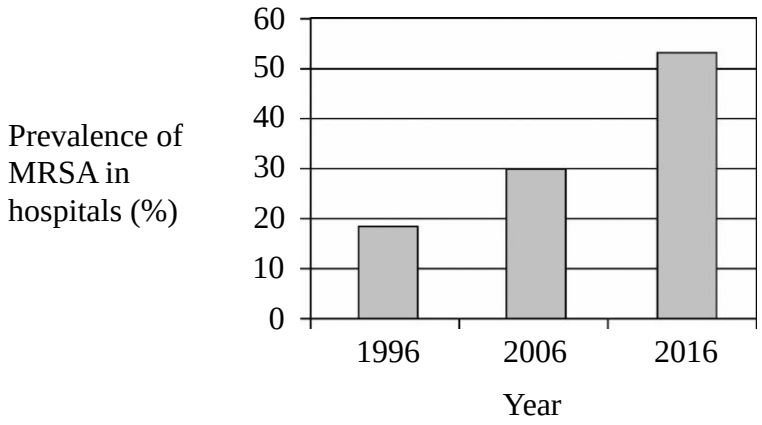
(b) After contracting typhoid, the person showed no symptoms of the disease for the first few days. Explain why this is especially dangerous if this person worked in a kitchen.

(2 marks)

(c) Suggest ***two*** precautions to prevent the spread of typhoid to other people. (2 marks)

(Total: 9 marks)

✂ 12. The graph below shows the prevalence of Methicillin-resistant *Staphylococcus aureus* (MRSA) in hospitals between 1996 and 2016. Methicillin is a type of antibiotic.



(a) What are antibiotics? (2 marks)

(b) Give one named example of antibiotics. (1 mark)

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(c) State ***two*** ways by which an antibiotic can affect *Staphylococcus aureus*. (2 marks)

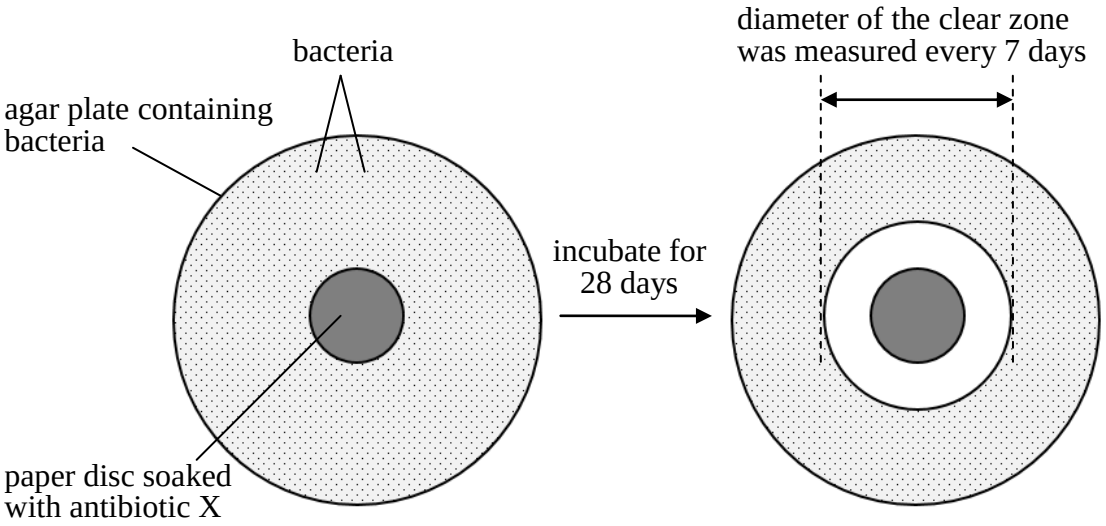
(d) Suggest an explanation to account for the change in the prevalence of MRSA in hospitals between 1996 and 2016. (5 marks)

(Total: 10 marks)

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C. Exam challenge (11 marks)

✂★ 13. The diagram below shows a way to test for the effectiveness of antibiotic X on three types of bacteria.



The table below shows the diameter of the clear zone over the 28 days.

Type of bacteria	Diameter of the clear zone (mm)			
	Day 7	Day 14	Day 21	Day 28
A	20.2	19.5	17.6	15.2
B	18.1	12.3	5.2	0
C	24.2	23.5	23.0	23.0

(a) Why was a clear zone formed around the paper disc? (2 marks)

(b) Describe and explain the result of bacteria A. (3 marks)

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(c) If you were a doctor, will you prescribe antibiotic X to a person infected by bacteria A, B or C? Explain your answer. (3 marks)

(d) A student would like to investigate the effect of different antibiotics on bacteria B. Suggest how he should amend the experiment to achieve this aim. (3 marks)

(Total: 11 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	D	2.	C	3.	D	4.	C	5.	B
6.	A	7.	A	8.	A	9.	B	10.	A

(1 mark each)

B. Problem solving (19 marks)

11. (a) (i) When a person have a high fever, his/her body temperature increases. (1)
The higher temperature may cause important metabolic enzymes to become denatured, (1)
hence the normal metabolism of the body is affected. (1)

(ii) Diarrhoea may lead to excess loss of mineral ions together with water. (1)
This would affect the water potential of the blood and tissue fluid. (1)

(b) This person has close contact with food and drinks. (1)
Bacteria can easily contaminate food and drinks during preparation if the person does not keep good hygiene practice. (1)

(c) Any ***two*** of the following: (2)

- washing hands after going to the toilet and before handling food (1)
- refrigerating food at the right temperature to inhibit the growth of *Salmonella typhi* (1)
- cooking food and boiling water thoroughly before consumption (1)

(Total: 9 marks)

12. (a) Antibiotics are chemicals produced by some microorganisms (1)
to kill or inhibit the growth of other microorganisms. (1)

(b) Penicillin / other acceptable answer (1)

(c) Any ***two*** of the following: (2)

- Inhibit replication of bacterial nucleic acid. (1)
- Inhibit the synthesis of proteins. (1)
- Destroy bacterial cell membranes. (1)
- Inhibit the synthesis of bacterial cell walls. (1)

(d) Methicillin is over-prescribed to treat *Staphylococcus aureus*. (1)
Genetic variations exist among *Staphylococcus aureus* in their resistance against methicillin. (1)
Methicillin only kills non-resistant form while the resistant form survives. (1)

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The resistant form continues to reproduce rapidly because of little intraspecific competition. (1)

The resistant form passes the methicillin-resistant trait to the next generation and *Staphylococcus aureus* without the Methicillin-resistant trait are gradually replaced by those with the resistant trait. (1)

(Total: 10 marks)

C. Exam challenge (11 marks)

13. (a) Antibiotic X diffused from the paper disc to the agar plate. (1)

Antibiotic X killed or inhibited the growth of bacteria, leaving a clear zone on the plate. (1)

(b) The diameter of the clear zone decreased over the 28 days. (1)

Some bacteria had developed resistance towards antibiotic X. (1)

More antibiotic X-resistant bacteria were reproduced and grew on the clear zone as the clear zone was free of bacteria and therefore with low competition stress for space and nutrients. (1)

(c) to a person infected by bacteria C (1)

The diameter of the clear zone formed on the agar plate with bacteria C is larger than those of bacteria A and B. This shows that antibiotic X can kill bacteria C more effectively. (1)

Bacteria C shows less resistance to antibiotic X. (1)

(d) Place paper discs soaked with different antibiotics on the surface of an agar plate inoculated with bacteria B. (1)

Measure and compare the clear zone formed by each of the antibiotics after incubating the agar plate for 1 to 2 days. (1)

The larger the diameter of the clear zone, the more effective the antibiotic is. (1)

(Total: 11 marks)

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